LD9 Final Project Summary

Final Documentation Package for the Bank Marketing Subscription Prediction Project

Prepared for: Ruturaj Mokashi

Learning Day 9 — Ensemble Methods Project Flow

What we completed from Notebook 1 to Notebook 6

1 Notebook 1 — Source Audit & Plan

- Audited the uploaded course files
- Built a coverage matrix for all required topics
- Selected the Bank Marketing dataset
- Defined the Day 9 notebook roadmap and evaluation plan

2 Notebook 2 — Data Preparation & Baselines

- Explored target balance, missing values, and feature types
- Created train/test split and preprocessing pipeline
- Trained baseline models: Dummy, Logistic Regression, Decision Tree, KNN
- Saved reusable data and baseline outputs

3 Notebook 3 — Bagging & Random Forest

- Compared Single Decision Tree vs Random Forest (100 trees)
- Reviewed feature importance for both models
- Tested n_estimators from 10 to 500 and found stagnation around 100
- Checked OOB score and Random Subspace Method via max_features
- Confirmed Random Forest clearly outperformed a single tree

4 Notebook 4 — Cross-Validation & StackingClassifier

- Ran 5-fold cross-validation for the base learners
- Built a StackingClassifier using RF, Logistic Regression, KNN, and Decision Tree
- Compared Logistic Regression vs Random Forest meta-models
- Studied model diversity and compared CV + held-out test results
- Best held-out stacking model: RF meta-model

5 Notebook 5 — Manual OOF Stacking & Blending

70% 30%

- Built manual 5-fold Out-of-Fold prediction features
- Trained a meta-model on OOF meta-features
- Built Blending with a 70/30 split
- Compared OOF Stacking vs Blending in performance and stability
- Found Manual OOF Stacking stronger than Blending

6 Notebook 6 — Three-Layer Stacking & Gradient Boosting

Level 2
Level 1
Level 0

- Created a 3-layer stacking system: Level 0, Level 1, Level 2
- Evaluated the final Three-Layer Stacking model
- Added a compact Gradient Boosting model for theory-to-practice coverage
- Compared complexity vs business value
- Practical recommendation: simplerr RF-meta stacking remained the best choice

Outcome So Far

- All major Day 9 ensemble topics from Notebook 1 to 6 were covered
- Practical methods included Bagging, Random Forest, OOF Stacking, Blending, Three-Layer Stacking, and Gradient Boosting
- Best practical model so far: StackingClassifier with Random Forest meta-model

Figure: High-level workflow from Notebook 1 to Notebook 6.

1. Project Objective

Build and compare ensemble learning models for the bank marketing dataset, covering Bagging, Random Forest, Stacking, Blending, Three-Layer Stacking, and Gradient Boosting, while fully satisfying the Learning Day 9 course scope and MVD requirements.

2. Dataset and Prediction Target

- Dataset: Bank Marketing dataset chosen for a business-relevant binary classification task.
- Target: Predict whether a customer subscribes to a term deposit.
- Project fit: Strong match to CRM, campaign optimization, BI reporting, and classification use cases.

3. Project Workflow

Notebook	Main focus	Main deliverable
1	Audit and planning	Coverage matrix and selected dataset
2	Data preparation and baselines	Preprocessing pipeline and baseline benchmark
3	Bagging and Random Forest	Tree vs forest comparison and RSM/OOB analysis
4	StackingClassifier	CV comparison and meta-model comparison
5	Manual OOF stacking and blending	Custom stacking logic and stability comparison
6	Three-layer stacking and boosting	Complexity-vs-value decision and boosting connection
7	Final proof notebook	MVD checklist and final recommendation

4. Best Final Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
StackingClassifier (RF meta)	0.8598	0.8176	0.9064	0.8597	0.9247
Three-Layer Stacking	0.8612	0.8369	0.8781	0.8570	0.9251
Manual OOF Stacking	0.8603	0.8354	0.8781	0.8562	0.9259
StackingClassifier (LR meta)	0.8589	0.8338	0.8771	0.8549	0.9265
Blending 70/30	0.8554	0.8314	0.8715	0.8509	0.9232
Gradient Boosting	0.8325	0.8109	0.8431	0.8267	0.9123

The recommended final model is the StackingClassifier with Random Forest meta-model because it gave the best overall balance for the business objective.

5. Important Findings

- Random Forest clearly outperformed a single unpruned Decision Tree.
- Feature importance results were broadly aligned between the tree and the forest, with duration, balance, age, day, and outcome/contact-related variables playing important roles.
- Random Forest performance improved quickly and then stagnated around 100 trees.
- Stacking improved over most individual base learners.
- Manual OOF stacking worked almost as well as automatic stacking and was valuable for learning the inner logic of stacking.
- Blending was simpler but slightly weaker and more split-dependent.
- Three-layer stacking was technically successful but not sufficiently better to justify the complexity.
- Gradient Boosting was useful conceptually but weaker than the best stacking models on this dataset.

6. Final Recommendation

Choose the RF-meta StackingClassifier for the final solution. It gives the strongest F1-score and the best recall, making it highly suitable for subscription targeting.

7. Limitations and Next Steps

- Threshold tuning was not the main focus; this could further improve business alignment.
- Explainability tools such as SHAP could be added later for richer model explanation.
- A production version would include monitoring, retraining schedule, and scoring pipeline automation.